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####

##sequoiadb####javascript#####

Sequoiadb javascript##

Sequoiadb#####JSON#####

#####". "(ASCII#0x2E)#####

```
{ name:"tom",address:{ street:{ street1:"1024 Wall Street",street2:"University Drive"},zipcode: 10000 } }
```

#####address####street####street1#####:

```
{ "address.street.street1":{ $exists : 1 } }
```

#####0## #1## #2#####

```
{ name: "tom", phone: [ "13175824", 1528345] }
```

#####:

```
{ "phone.0 ":"13175824" }
```

#####

###	##	##
db.createCS()	#####	db.createCS("foo")
db.dropCS()	#####	db.dropCS("foo")
db.getCS()	#####	db.getCS("foo")
db.createRG()	#####	db.createRG("group")
db.getRG()	#####	db.getRG("group")
db.list()	####	db.list()
db.listCollectionSpaces()	#####	db.listCollectionSpaces()
db.listCollections()	####	db.listCollections()
db.listReplicaGroups()	#####	db.listReplicaGroups()
db.snapshot()	####	db.snapshot(0)
db.resetSnapshot()	####	db.resetSnapshot(0)
db.startRG()	#####	db.startRG("groupName")
db.createUsr()	#####	db.createUsr("root","admin")
db.dropUsr()	#####	db.dropUsr("root","admin")
db.transBegin()	####	db.transBegin()
db.transCommit()	####	db.transCommit()
db.transRollback()	####	db.transRollback()

#####

###	##	##
db.collectionspace.createCL()	####	db.foo.createCL("bar")
db.collectionspace.dropCL()	####	db.foo.dropCL("bar")
db.collectionspace.getCL()	#####	db.foo.getCL("bar")

####

###	##	##
db.collectionspace.collection.insert()	#####	db.foo.bar.insert({name:"Tom",age:20,[123,345]})
db.collectionspace.collection.count()	#####	db.foo.bar.count({age:{>:20}})
db.collectionspace.collection.find()	####	db.foo.bar.find()
db.collectionspace.collection.remove()	####	db.foo.bar.remove()
db.collectionspace.collection.createIndex()	####	db.foo.bar.createIndex("myIndex", {age:-1}, false)
db.collectionspace.collection.dropIndex()	####	db.foo.bar.dropIndex("myIndex")
db.collectionspace.collection.getIndex()	#####	db.foo.bar.getIndex("myIndex")
db.collectionspace.collection.listIndexes()	####	db.foo.bar.listIndexes()
db.collectionspace.collection.update()	####	db.foo.bar.update({\$set:{age:25}}, {age:{<:20}}, {"":"myIndex"})
db.collectionspace.collection.update()	####	db.foo.bar.update({\$set:{age:25}}, {a:1}, {"":"myIndex"})
db.collectionspace.collection.split()	####	db.foo.bar.split("group1","group2", {age:20})

####

###	##	##
rg.start()	#####	rg.start()
rg.getDetail()	#####	rg.getDetail()
rg.createNode()	####	rg.createNode(<hostname>,<port>,<path>)
rg.getMaster()	#####	rg.getMaster()
rg.getSlave()	#####	rg.getSlave()
rg.getNode()	#####	rg.getNode()
rg.stop()	#####	rg.stop()

6#|#0penTopic#|#####

####

###	##	##
node.start()	####	node.start()
node.connect()	#####	node.connect()
node.getHostName()	#####	node.getHostName()
node.getServiceName()	#####	node.getServiceName()
node.getNodeDetail()	#####	node.getNodeDetail()
node.stop()	####	node.stop()

Cursor##

###	##	##
cursor.sort()	#####	db.foo.bar.find().sort({age:1})
cursor.hint()	#####	db.foo.bar.find().hint({"":"Index"})
cursor.limit()	#####	db.foo.bar.find().limit(10)
cursor.skip()	#####	db.foo.bar.find().skip(10)
cursor.current()	#####	db.foo.bar.find().current()
cursor.deleteCurrent()	#####	db.foo.bar.find().deleteCurrent()
cursor.next()	#####	db.foo.bar.find().next()
cursor.updateCurrent()	#####	db.foo.bar.find().updateCurrent({\$set:{age:25}})
cursor.size()	#####	db.foo.bar.find().size()
cursor.toArray()	#####	db.foo.bar.find().toArray()

db.createCS()

db.createCS(<name>,[pageSize])

#####

####

###	####	##	####
name	string	#####	#
pageSize	int(32)	#####4096B	#

##

#####name#pageSize####name#####pageSize#32#int###

("<####>",[pagesize])



Note:

```
# name#####(.)#####($)#####127B#
# #####
# ##### 4096B #
```

##

#####foo#####

db.createCS("foo")

#####foo#####1024B

db.createCS("foo", 1024)

db.dropCS()

db.dropCS(<name>)

#####

####

###	####	##	####
name	string	##### #####	#

##

#####name###name###string#####

("<#####>")



Note:

```
# name#####(.)#####($)#####127B#
# #####.
```

##

#####foo#####foo###

db.dropCS("foo")

db.getCS()

db.getCS(<name>)

#####

####

###	####	##	####
name	string	##### #####	#

8#|#0penTopic#|#####

```
##
getCS() #####name###name##### #
("<#####>")
```

Note:

name#####(.)#####(\$)#####127B#
#####

```
##
# #####foo#####foo####
db.getCS(#foo#)
```

db.createRG()

```
db.createRG(<groupname>)
#####GroupId
####
```

###	####	##	####
groupname	string	##### #####	#

```
##
createRG() #####groupname###groupname#####
("<####>")
```

Note:

#####(.)#####(\$)#####127B#

#####group#####
db.createRG(#group#)

db.getRG()

```
db.getRG(##)
#####
####
```

###	####	##	####
name	string	##### #####	name#id##
id	int	###id#####	name#id##

##

```
getRG()#####name#id###name#####id#int## ##### id #####
("<####>"|<id>)
```



Note:

```
# name#####.(.)#####($)#####127B#
```

##

```
# ##name#####group###
```

```
db.getRG("group")
```

```
# ##id#####group###
```

```
db.getRG("1000")
```

db.list()

```
db.list(<listType>,[con],[sel],[sort])
```

```
#####
```

####

###	####	##	####
listType	##	#####0#7	#
con	json##	#####con#### #####null#####	#
sel	json##	#####null#### #####	#
sort	json##	#####1# ###-1###	#

##

```
list()#####listType#con#sel#sort#####listType#####json#####
```

```
{"listType":"<####>","con":{"###1":{"###1":"#1"},"###2":{"###2":"#2"}...},"sel":{"###1":"", "###2":"",...},"sort":{"###1":-1|1,"###2":1|-1,...}}}
```



Note:

```
# listType####0#7
```

```
# sel#####json#####{"###1":"#1","###2":"#2",...}#####  
#####{"###1":"#1","###2":"#2",...}
```

```
# #####".#####(")#
```

##

```
# ##listType###0#
```

```
db.list(0)
```

```
###
```

```
{  
  "SessionID": 4,
```

10#|#OpenTopic#|#####

```
    "Contexts": [ 0 ]
  } ...
# ##listType##6#
db.list(6)

###
{
  "Name": "foo",
  "ID": 4094,
  "Logical ID": 1,
  "PageSize": 4096,
  "Sequence": 1,
  "NumCollections": 1,
  "CollectionHWM": 3,
  "Size": 172032000
}
# db.list(6, {"Logical ID": {$gt:1}}, {Name:"space", ID:2}, {Name#1})
#####Logical ID##1#####Name#ID#####Name#####
{
  "ID": 4091,
  "Name": "foo"
}
{
  "ID": 4093,
  "Name": "name"
}...
```

db.listCollectionSpaces()

```
db.listCollectionSpaces()
#####

##
# db.listCollectionSpace()

###
{
  "Name": "foo"
}...
```

db.listCollections()

```
db.listCollections()
#####

##
# db.listCollections()

###
{
  "Name": "foo.bar",
  "Details": [
    {
      "ID": 0,
      "Logical ID": 1,
```

```

        "Sequence": 1,
        "Indexes": 2,
        "Status": "Normal"
      }
    ]
  }
}

```

db.listReplicaGroups()

```
db.listReplicaGroups()
```

```
#####
```

```
##
```

```
# #####
```

```

> db.listReplicaGroups()
###{
  "Group": [
    {
      "dbpath": "/opt/sequoiadb/data/60000",
      "HostName": "vmsvr2-suse-x64",
      "Service": [
        {
          "Type": 0,
          "Name": "60000"
        },
        {
          "Type": 1,
          "Name": "60001"
        },
        {
          "Type": 2,
          "Name": "60002"
        },
        {
          "Type": 3,
          "Name": "60003"
        }
      ],
      "NodeID": 1000
    },
    {
      "dbpath": "/opt/sequoiadb/data/60010",
      "HostName": "vmsvr2-suse-x64",
      "Service": [
        {
          "Type": 0,
          "Name": "60010"
        },
        {
          "Type": 1,
          "Name": "60011"
        },
        {
          "Type": 2,
          "Name": "60012"
        },
        {
          "Type": 3,
          "Name": "60013"
        }
      ]
    }
  ]
}

```

12#|#0penTopic#|#####

```
    }
  ],
  "NodeID": 1001
}
],
"GroupID": 1001,
"GroupName": "group",
"PrimaryNode": 1001,
"Role": 0,
"Status": 1,
"Version": 5,
"_id": {
  "$oid": "517b2fc33d7e6f820fc0eb57"
}
}
```

#####60000#60010###60010#####

db.snapshot()

db.snapshot(<snapType>,[con],[sel],[sort])

#####

####

###	####	##	####
snapType	##	#####0#7	#
con	json##	#####con#### #####null#####	#
sel	json##	#####null#### #####	#
sort	json##	#####1# ###-1###	#

##

snapshot()#####listType#con#sel#sort####listType#####json#####

```
{"snapType": "<####>","con": {"###1": {"###1": "#1"}, "###2": {"###2": "#2"}...}, "sel": {"###1": "#1", "###2": "#2",...}, "sort": {"Order:-1|1"}]}
```



Note:

```
# snapType#####0#7#
# sel#####json#####{"###1": "#1", "###2": "#2",...}#####
# #####{"###1": "#1", "###2": "#2",...}#
# #####".#####(")#
```

##

##snapType###0#

db.snapshot(0)

#####

```
{
  "SessionID": 3,
```

```

    "Contexts": [
      {
        "ContextID": 1,
        "DataRead": 0,
        "IndexRead": 0,
        "QueryTimeSpent": 0,
        "StartTimestamp": "2013-04-08-17.17.02.135620"
      }
    ]
  }
}

# db.snapshot(0, {"Contexts.0.ContextID": {$gte:0}},
# {SessionID:"","Contexts.0.StartTimestamp":""}, {SessionID#1})

#####Contexts.0.ContextID##0## (#. "#####)#####SessionID#Contexts.StartTimestamp##
#####

###SessionID####:
{
  "Contexts": [
    {
      "StartTimestamp": "2013-04-09-09.31.03.270965"
    }
  ],
  "SessionID": 5
}

```

db.resetSnapshot()

```
db.resetSnapshot([cond])
```

```
####
```

```
####
```

###	####	##	####
con	json##	#####con#### #####null#####	#

```
##
```

```
resetSnapshot()#####con#####json###
```

```
{["con": {"###1": {"###1": "#1"}, "###2": {"###2": "#2"}...}]}
```

```
##
```

```
# ##SessionID##1####
```

```
db.resetSnapshot({SessionID: {$gt:1}})
```

db.startRG()

```
db.startRG(<name>)
```

```
#####rg.start()#
```

####

###	####	##	####
name	sring	#####	#

##

db.startRG()#####name#####

("<####>")



Note:

#####

##

#####group#####

db.startRG("group")

db.createUser()

db.createUser(<name>,<password>)

#####

####

###	####	##	####
name	string	###	#
password	string	##	#

##

createUser() #####name#password#####

("<###>","<##>")



Note:

#####

#####

db = new Sdb("<hostname>","<port>","<name>","<password>")

##

#####root####admin#####

db.createUser("root","admin")

db.dropUser()

db.dropUser(<name>,<password>)

#####

####

###	####	##	####
name	string	###	#
password	string	##	#

##

dropUsr() #####name#password#####

("<###>", "<##>")

##

#####root####admin#####

db.dropUsr("root", "admin")

db.transBegin()

db.transBegin()

#####sequoiaDB#####

##

#####

db.transBegin()

db.transCommit()

db.transCommit()

#####

##

#####

db.transCommit()

db.transRollback()

db.transRollback()

#####

##

#####

db.transRollback()

db.collectionspace.createCL()

db.collectionspace.createCL(<name>, [shardingkey])

#####Collection## #####

16#|#OpenTopic#|#####

####

###	####	##	####
name	string	##### #### #	#
shardingkey	json##	#### ##### ##### #####	#

##

createCL()#####name#shardingkey##### name ##### ####shardingkey###JSON###

```
{"name": "<###>", ["shardingkey": {"###1": 1|-1, "###2": 1|-1, ...}]}
```

 Note:

```
# shardingkey###JSON###JSON#####1##-1#####  
# name #####. #####$#####127B#####
```

##

#####foo####bar#####

```
db.foo.createCL("bar")
```

#####foo####bar####age#####

```
db.foo.createCL("bar", {ShardingKey: {"age": 1}})
```

db.collectionspace.dropCL()

db.collectionspace.dropCL(<name>)

#####

####

###	####	##	####
name	string	#### ##### #### #	#

##

dropCL() ##### name ##### name #####

```
{"name": "<###>"}
```

 Note:

```
# name #####. #####$#####127B#####  
# #####
```

##

#####foo####bar#####

```
db.foo.dropCL("bar")
```


db.collectionspace.getCL()

```
db.collectionspace.getCL(<name>)
```

```
#####
```

```
####
```

###	####	##	####
name	string	#### ##### ##### #	#

```
##
```

```
getCL() ##### name ##### name #####
```

```
{"name": "<###>"}
```



Note:

```
# name #####.#####$#####127B#####
# #####
```

```
##
```

```
# #####foo###bar#####
```

```
db.foo.getCL("bar")
```

db.collectionspace.collection.insert()

```
db.collectionspace.collection.insert(<doc|docs>, [flag])
```

```
#####db.createCS("foo")#####foo#
```

```
#####db.foo.createCL("bar")#####foo#####bar#####bar#####
```

```
####
```

###	####	##	####
doc docs	json##	#####doc#### ##docs#####	#
flag	jsBool	#####_id##### 0##1,####0#####1 ####_id#	#

```
##
```

```
insert() ##### doc|docs # flag #### , doc|docs ### JSON ## #flag#JavaScript#Boolean#####0#  
#1#
```

```
doc#
```

```
{ { "< ### 1>": "< # >", "< ### 2>": "< # >", #} [, "flag":<0|1>] }
```

```
docs#
```

```
{ [
```

18#|#0penTopic#|#####

```
{ { "< ### 1>": "< # >", "< ### 2>": "< # >", # } [, "flag": <0|1> ] },  
[ { "< ### 1>": "< # >", "< ### 2>": "< # >", # } [, "flag": <0|1> ] }, ...  
] }
```



Note:

#####_id###sequoiadb#####_id#####

##

###_id#####

```
db.foo.bar.insert({name:"Tom",age:20})
```

#####bar#####name#####Tom##age#####20#_id#####

```
{ "_id": { "$oid": "515152ba49af395200000000" }, "name": "Tom", "age": 20 }
```

#####_id#####

```
db.foo.bar.insert({_id:10,age:20})
```

#####bar#####_id#####10#age#####20#

```
{ "_id": 10, "age": 20 }
```

#####

```
db.foo.bar.insert([ {_id:20,name:#Mike#,age:15}, {name:#John#,age:25,phone:123} ])
```

#####bar#####

1)#####_id###20###name###"Mike"###age###15#

2)#####_id#####name###"John"###age###25###phone###123#

```
{  
  "_id": 20,  
  "name": "Mike",  
  "age": 15  
}
```

```
{  
  "_id": { "$oid": "5151557a49af395200000001" },  
  "name": "John",  
  "age": 25,  
  "phone": 123  
}
```

db.collectionspace.collection.count()

db.collectionspace.collection.count([cond])

#####

####

###	####	##	####
cond	json##	##### ##### #####	#

##

count() #####cond#####JSON###

```
[ [{"###1": {"###1": "#1"}, "###2": {"###2": "#2"}, ... } ]
```

##

#####bar#####cond

db.foo.bar.count()

#####name#####"Tom"#age#####25####

db.foo.bar.count({name:#Tom#,age:{\$gt:25}})

db.collectionspace.collection.find()

db.collectionspace.collection.find([cond],[sel])

#####(cursor)##Sequoiadb#####

####

###	####	##	####
cond	json##	##### #####	#
sel	json##	##### ##### #####	#

##

find()#####cond#sel#####JSON#####cond#####sel#####

{[{"###1":{"###1":"#1","###2":{"###2":"#2"},...}], [{"###1":"", "###2":"",..}]}



Note:

sel#####json#####{"###1":"#1","###2":"#2",...}#####

#####{"###1":"#1","###2":"#2",...}#

##

#####cond # sel

db.foo.bar.find()

#####cond#####

db.foo.bar.find({age:{\$gt:25},name:"Tom"})

#####bar#####age#####25#name#####"Tom"###

#####sel#####{age:25,type:"system"}#{age:20,name:"Tom",type:"normal"}

db.foo.bar.find(null,{age:"",name:""})

#####age###name#####{age:25,name:""}#{age:20,name:"Tom"}#####name#####
name:""#

db.collectionspace.collection.remove()

db.collectionspace.collection.remove([cond],[hint])

#####

####

###	####	##	####
cond	json##	##### #####	#
hint	json##	#####	#

##

```
cond#####json#####(##{} )#####hint#####json#####  
#####null#####  
{[{"###1":{"###1":"#1","###2":{"###2":"#2"},...}], [{"":"###" | null}]}
```

##

```
# #####  
db.foo.bar.remove()  
# #####cond#####  
db.foo.bar.remove({age: {$gte:20}}, {"":"myIndex"})  
#####"myIndex"#####age#####20###
```

db.collectionspace.collection.createIndex()

```
db.collectionspace.collection.createIndex(<name>,<indexDef>,[isUnique])  
#####
```

####

###	####	##	####
name	string	##### ###	#
indexDef	json##	##### #####1#### #####-1#####	#
isUnique	Boolean	#####false### #true#####	#

##

```
createIndex()#####name#indexDef#isUnique##### ##name#####indexDef####JSON### indexDef####  
#####1##-1###1#####-1####isUnique#####false#  
{"name": "<###>","indexDef": {"<###1>":<1|-1> [,<###2>":<1|-1>... ] },["isUnique":<true|false>]}
```



Note:

```
# #####  
# #####(. )#####($)#####127B#
```

```
##
# #####bar#####ageIndex#####age#####
db.foo.bar.createIndex("ageIndex", {age:1}, false)
```

db.collectionspace.collection.dropIndex()

```
db.collectionspace.collection.dropIndex(<name>)
#####
####
```

###	####	##	####
name	string	##### ###	#

```
##
dropIndex()#####name#####name#####
{"name": "<###>"}
```

Note:

# #####
#####(.)#####(\$)#####127B#

```
##
# #####bar###ageIndex#####ageIndex####
db.foo.bar.dropIndex("ageIndex")
```

db.snapshot()

```
db.snapshot(<snapType>, [con], [sel], [sort])
#####
####
```

###	####	##	####
snapType	##	#####0#7	#
con	json##	#####con#### #####null#####	#
sel	json##	#####null#### #####	#
sort	json##	#####1# ###-1###	#

```
##
snapshot()#####listType#con#sel#sort####listType#####json#####
{"snapType":"<####>",[{"con":{"###1":{"###1":"#1"},"###2":{"###2":"#2"}...}],["sel":{"###1":"","###2":"","..."}],["sort":{"Order:-1|1}"}]}
```

Note:

```
# snapType####0#7#
# sel#####json#####{"###1":"#1","###2":"#2",...}#####
#####{"###1":"#1","###2":"#2",...}#
# #####".#####(")#
```

```
##
# ##snapType##0#
db.snapshot(0)
```

```
#####
{
  "SessionID": 3,
  "Contexts": [
    {
      "ContextID": 1,
      "DataRead": 0,
      "IndexRead": 0,
      "QueryTimeSpent": 0,
      "StartTimestamp": "2013-04-08-17. 17. 02. 135620"
    }
  ]
}
```

```
# db.snapshot(0, {"Contexts.0.ContextID": {$gte:0}},
{SessionID:"", "Contexts.0.StartTimestamp":""}, {SessionID#1})
```

```
#####Contexts.0.ContextID##0##(#.#####)#####SessionID#Contexts.StartTimestamp##
####
```

```
###SessionID####:
{
  "Contexts": [
    {
      "StartTimestamp": "2013-04-09-09. 31. 03. 270965"
    }
  ],
  "SessionID": 5
}
```

db.collection.space.collection.getIndex()

```
db.collection.space.collection.getIndex(<name>)
#####
```

####

###	####	##	####
name	strsing	##### ###	#

##

getIndex() #####name####name#####

```
{"name": "<###>"}
```



Note:

#####

#####(.)#####(\$)#####127B#

##

#####bar###ageIndex#####ageIndex####

```
db.foo.bar.getIndex("ageIndex")
```

db.collectionspace.collection.listIndexes()

db.collectionspace.collection.listIndexes()

#####

##

#####bar#####

```
db.foo.bar.listIndexes()
```

db.collectionspace.collection.update()

db.collectionspace.collection.update(<rule>, [cond], [hint])

#####

####

###	####	##	####
rule	json##	#####rule#### ##	#
cond	json##	##### #####	#
hint	json##	#####	#

##

update() #####rule###rule###json###cond#hint#####hint#####

###json#####null#####

#####{"":null}##{"": "<indexname>"}

```
<<{"###1": {###1: "#"}, "###2": {"###2": "#2"}, ...}>, [{"###1": {"###1": "#1"}, "###2": {"###2": "#2"}, ...}], [{"": "###" | null}]}
```

##

#####rule#####cond#hint#####

```
db.foo.bar.update({$inc: {age:1}})
```

```
#####bar##age###age#####1#
# #####rule#cond###
db.foo.bar.update({$unset:{age:""},{age:{$exists:1},name:{$exists:0}})
#####bar##age#####name#####age#####
# #####
db.foo.bar.update({$inc:{age:1}},{age:{$gt:20}},{":testIndex"})>
#####testIndex###bar#age#####20#####age#####1#
```

db.collectionspace.collection.upsert()

```
db.collectionspace.collection.upsert(<rule>,[cond],[hint])
#####upsert###update#####cond#####update
#####upsert#####
####
```

###	####	##	####
rule	json##	#####rule#### ##	#
cond	json##	##### #####	#
hint	json##	#####	#

```
##
upsert()#####rule###rule###json###cond#hint#####hint#####
###json#####null#####
#####{"":null}##{"":<indexname>}
{<{"###1":{"###1":"#"},"###2":{"###2":"#2"},...}>,[{"###1":{"###1":"#1"},"###2":{"###2":"#2"},...}],[{"":"###"|null}]}
```

```
##
####bar#####
```

```
{
  "_id": {
    "$oid": "516a76a1c9565daf06030000"
  },
  "age": 10,
  "name": "Tom"
}
{
  "_id": {
    "$oid": "516a76a1c9565daf06050000"
  },
  "a": 10,
  "age": 21
}
```


#####rule#####cond#hint#####

```
db.foo.bar.upsert({$inc:{age:1},$set:{name:"Mike"}})
```

#####update#####bar#####age####1#name#####Mike"####

#name#####\$set####name#####find#####

```
{
  "_id": {
    "$oid": "516a76a1c9565daf06030000"
  },
  "age": 11,
  "name": "Mike"
}
{
  "_id": {
    "$oid": "516a76a1c9565daf06050000"
  },
  "a": 10,
  "age": 22,
  "name": "Mike"
}
```

#####rule#cond###

```
db.foo.bar.upsert({$inc:{age:3}}, {type:{$exists:1}})
```

#####bar###type#####age####3#####type###

###upsert#####_id###age####_id#####age####3#

```
{
  "_id": {
    "$oid": "516a76a1c9565daf06030000"
  },
  "age": 11,
  "name": "Mike"
}
{
  "_id": {
    "$oid": "516a76a1c9565daf06050000"
  },
  "a": 10,
  "age": 22,
  "name": "Mike"
}
{
  "_id": {
    "$oid": "516cfc334630a7f338c169b0"
  },
  "age": 3
}
```

#####testIndex#

```
db.foo.bar.upsert({$inc:{age:1}}, {age:{$gt:20}}, {"":"testIndex"})>
```

#####update#####testIndex####bar#age#####20#####age####1#####

```
{
  "_id": {
    "$oid": "516a76a1c9565daf06050000"
  },
  "a": 10,
  "age": 23,
  "name": "Mike"
}
```

```
rg.start()

rg.start()
#####db.startRG(<name>)db.startRG(<name>)#####

##
# #####
    rg.start() //###db.startRG("rg")
```

```
rg.createNode()

rg.createNode(<HostName>,<Port>,<Path>)
#####
👉 Note:
    #####
```

####

###	####	##	####
HostName	string	###	#
Port	string	#####	#
Path	string	####	#

```
##
rg.createNode()#####HostName#Port#Path#####
("<###>","<###>","<###>")

##
# #####rg#####60000###
    rg.createNode("vmsvr2-suse-x64",60000,"/opt/sequoiadb/data/60000")
```

```
👉 Note:
    #####5#####5#####
```

```
rg.getDetail()

rg.getDetail()
#####

##
# > rg.getDetail()
  {
    "Group": [
      {
        "dbpath": "/opt/sequoiadb/data/60000",
        "HostName": "vmsvr2-suse-x64",
```

```

    "Service": [
      {
        "Type": 0,
        "Name": "60000"
      },
      {
        "Type": 1,
        "Name": "60001"
      },
      {
        "Type": 2,
        "Name": "60002"
      },
      {
        "Type": 3,
        "Name": "60003"
      }
    ],
    "NodeID": 1000
  },
  {
    "dbpath": "/opt/sequoiadb/data/60010",
    "HostName": "vmsvr2-suse-x64",
    "Service": [
      {
        "Type": 0,
        "Name": "60010"
      },
      {
        "Type": 1,
        "Name": "60011"
      },
      {
        "Type": 2,
        "Name": "60012"
      },
      {
        "Type": 3,
        "Name": "60013"
      }
    ],
    "NodeID": 1001
  }
],
"GroupID": 1001,
"GroupName": "group",
"PrimaryNode": 1001,
"Role": 0,
"Status": 1,
"Version": 3,
"_id": {
  "$oid": "517b2fc33d7e6f820fc0eb57"
}
}

```

```
rg.getMaster()
```

```
rg.getMaster()
```

```
#####
```

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##

#####

```
> rg.getMaster()
###vmsvr2-suse-x64:60010
```

rg.getSlave()

rg.getSlave()

#####

##

#####

```
> rg.getSlave()
###vmsvr2-suse-x64:70010
```

rg.getNode()

rg.getNode(<nodename>|<hostname>,<servicename>)

#####

####

###	####	##	####
nodename	string	####	nodename#hostname ###
hostname	string	###	nodename#hostname ###
servicename	string	#####	#

##

rg.getNode()#####

```
("<####>|<###>","<#####>")
```

##

#####

```
> rg.getNode("vmsvr2-suse-x64","70000")
###vmsvr2-suse-x64:70000
```

rg.stop()

rg.stop()

#####

##

#####

```
rg.stop()
```

node.start()

```
node.start()
#####
```

```
##
```

```
# #####node### #
node.start()
```

node.connect()

```
node.connect()
#####node.connect().help()#####
```

```
##
```

```
# #####node#
> node.connect()
###vmsvr2-suse-x64:60000
```

node.getHostName()

```
node.getHostName()
#####
```

```
##
```

```
# #####node####
> node.getHostName()
###vmsvr2-suse-x64
```

node.getServiceName()

```
node.getServiceName()
#####
```

```
##
```

```
# #####node#####
> node.getServiceName()
###60000
```

node.getNodeDetail()

```
node.getNodeDetail()
#####
```

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##

#####node###

```
> node.getNodeDetail()  
###1000:vmsvr2-suse-x64:60000(group)  
##"1000"###ID#NodeID##"vmsvr2-suse-x64"#####HostName##  
"60000"#####ServiceName##"(group)"#####
```

node.stop()

node.stop()

#####

##

#####node###

```
node.stop()
```

cousor.sort()

cousor.sort(<sort>)

#####

####

###	####	##	####
sort	json##	##### ##1##-1,1#####-1 #####	#

##

cousor.sort()#####sort#####json#####sort#####

##sort#####

```
{<###1>:<-1|1>,<###2>:<-1|1>,...}
```

##

#####bar#age#####20#####name#age#####age#####

```
db.foo.bar.find({age:{$gt:20}}, {age:" ",name:" "}).sort({age:1})
```



Note:

##find()#####age#name#####

sort()#####age#name#####find###sel#####

#sort()#####

cursor.hint()

cursor.hint(<hint>)

#####

####

###	####	##	####
hint	json##	#####	#

##

```
cursor.hint()#####hint#####hint#####hint#####
```

```
###json#####null#####
```

####

```
{"":null}##{"":<indexname>}
```

##

```
# ####ageIndex####bar###age#####
```

```
db.foo.bar.find({age:{$exists:1}}).hint({"": "ageIndex"})
```

cursor.limit()

```
cursor.limit(<num>)
```

```
#####
```

####

###	####	##	####
num	int	#####	#

##

```
cousor.limit()#####num####int#####num#####
```

```
###5#####num###5#
```

##

```
# ####bar#age#####10#####10####
```

```
db.foo.bar.find({age:{$gt:10}}).limit(10)
```



Note:

```
#####10#####10#####10##
```

cursor.skip()

```
cursor.skip(<num>)
```

```
#####
```

####

###	####	##	####
num	int	#####	#

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##

```
cousor.skip()#####num####int#####num#####num###0#####  
#####3#####num###2#
```

##

```
# #####bar#age#####10#####5#####  
db.foo.bar.find({age:{$gt:10}}).skip(4)
```



Note:

```
#####5#####5###5####
```

cursor.current()

```
cursor.current()  
#####cursor.next()###
```

##

```
# #####bar#age##10#####  
db.foo.bar.find({age:{$gt:10}}).current()
```

cursor.deleteCurrent()

```
cursor.deleteCurrent()  
#####  
##  
# #####bar###age#####10#####  
db.foo.bar.find({age:{$gt:10}}).deleteCurrent()
```

cursor.next()

```
cursor.next()  
#####cursor.current()###  
##  
# #####bar#age##10#####  
db.foo.bar.find({age:{$gt:10}}).next()
```

cursor.updateCurrent()

```
cursor.updateCurrent(<rule>)  
#####
```


####

###	####	##	####
rule	json##	#####rule#### ##	#

##

```
cursor.updateCurrent()#####rule#####json###
```

#####

```
{"###1":{"###1":"#1","###2":{"##2":"#2"},...}
```

##

```
# #####bar###age#####10#####age#####
```

```
db.foo.bar.find({age:{$gt:10}}).updateCurrent({$inc:{age:5}})
```

```
cursor.size()
```

```
cursor.size()
```

#####

##

```
# #####bar#age##10#####
```

```
db.foo.bar.find({age:{$gt:10}}).size()
```

```
cursor.toArray()
```

```
cursor.toArray()
```

#####

##

```
# #####bar#age##5#####
```

```
db.foo.bar.find({age:{$gt:10}}).size()
```

```
###{
  "_id": {
    "$oid": "516a76alc9565daf06030000"
  },
  "age": 10,
  "name": "Tom"
}, {
  "_id": {
    "$oid": "516a76alc9565daf06050000"
  },
  "age": 20,
  "a": 10
}, {
  "_id": {
    "$oid": "516a76alc9565daf06040000"
  },
  "age": 15
```

}

###

####

###	##	##
\$gt	##	db. foo. bar. find({age: {\$gt:20}})
\$gte	####	db. foo. bar. find({age: {\$gte:20}})
\$lt	##	foo. bar. find({age: {\$lt:20}})
\$lte	####	db. foo. bar. find({age: {\$lte:20}})
\$ne	###	db. foo. bar. find({age: {\$ne:20}})
\$et	##	db. foo. bar. find({age: {\$et:20}})
\$in	#####	db. foo. bar. find({age: {\$in:[20,21]}})
\$nin	#####	db. foo. bar. find({age: {\$nin:[20,21]}})
\$all	##	db. foo. bar. find({age: {\$all:[20,21]}})
\$and	#	db. foo. bar. find({\$and: {age:20, name:"Tom"}})
\$not	#	db. foo. bar. find({\$not: {age:20, name:"Tom"}})
\$or	#	db. foo. bar. find({\$or: {age:20, name:"Tom"}})
\$type	####	db. foo. bar. find({age: {\$type:16}})
\$exists	##	db. foo. bar. find({age: {\$exists:1}})
\$elemMatch	####	db. foo. bar. find({age: {\$elemMatch:20}})
\$size	##	db. foo. bar. find({array: {\$size:3}})
\$regex	#####	db. foo. bar. find({str: {\$regex:'dh.*fj', \$options:'i'}})

###	##	##
\$options	#####	db.foo.bar.find({str: {\$regex:'dh.*fj', \$options:'i'}})

####

###	##	##	##
\$inc	##	#####	db.foo.bar.update({\$inc: {age:25}})
\$set	#####	#####	db.foo.bar.update({\$set: {age:10}})
\$unset	#####	#####	db.foo.bar.update({\$unset: {age:""}})
\$addto	####	##### #####	db.foo.bar.update({\$addto: {array:[3,4,5]}})
\$pop	#####N ###	##### N ##### ##### N ## 0 ##### -N # ### #	db.foo.bar.update({\$pop: {array:2}})
\$pull	####	##### #####	db.foo.bar.update({\$pull: {array:2}})
\$pull_all	####	##### ##### #	db.foo.bar.update({\$pull_all: {array:[2,3,4]}})
\$push	####	##### ### #	db.foo.bar.update({\$push: {array:2}})
\$push_all	####	##### #####	db.foo.bar.update({\$push_all: {array:[2,3,4]}})



Note:

#sequoiadb#####-#####JSON{#####}#### #####\$set###

##

db.collectionspace.collection.find({a:42}) ### db.collectionspace.collection.find({a:
{\$et:42}})

#####a##42###

###

###	##	##
\$gt	##	db.foo.bar.find({age: {\$gt:20}})
\$gte	####	db.foo.bar.find({age: {\$gte:20}})

###	##	##
\$lt	##	foo. bar. find({age: { \$lt: 20}})
\$lte	####	db. foo. bar. find({age: { \$lte: 20}})
\$ne	###	db. foo. bar. find({age: { \$ne: 20}})
\$in	#####	db. foo. bar. find({age: { \$in: [20, 21]}})
\$nin	#####	db. foo. bar. find({age: { \$nin: [20, 21]}})
\$all	##	db. foo. bar. find({age: { \$all: [20, 21]}})
\$and	#	db. foo. bar. find({ \$and: {age: 20, name: "Tom"}})
\$not	#	db. foo. bar. find({ \$not: {age: 20, name: "Tom"}})
\$or	#	db. foo. bar. find({ \$or: {age: 20, name: "Tom"}})
\$type	####	db. foo. bar. find({age: { \$type: 16}})
\$exists	##	db. foo. bar. find({age: { \$exists: 1}})
\$elemMatch	####	db. foo. bar. find({age: { \$elemMatch: 20}})
\$size	##	db. foo. bar. find({array: { \$size: 3}})
\$regex	#####	db. foo. bar. find({str: { \$regex: 'dh.*fj', \$options: 'i'}})
\$options	#####	db. foo. bar. find({str: { \$regex: 'dh.*fj', \$options: 'i'}})

\$gt

##

{<###>#{\$gt:<#>}}

##

\$gt####"###"####(>)##"#"####

##

#####bar#age#####20####

db.foo.bar.find({age:{\$gt:20}})

\$gt#####update()#####service##ID###

##2#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.ID":{\$gt:2}})

\$gte

##

{<###>#{\$gte:<#>}}

##

\$gte#####"###"#####(>=)##"#"####

##

#####foo###bar#####age#####20####

db.foo.bar.find({age:{\$gte:20}})

\$gte#####update()#####service##ID#####

##2#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.ID":{\$gte:2}})

\$lt

##

{<###>#{\$lt:<#>}}

##

\$gt#####"###"#####(<)##"#"####

##

#####bar#####age#####20####

db.foo.bar.find({age:{\$lt:20}})

\$lt#####update()#####service##ID#####

2#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.ID":{\$lt:15}})

\$lte

##

{<###>#{\$lte:<#>}}

##

\$lte#####"###"#####(<=)##"#"####

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##

#####foo###bar#####age#####20####

db.foo.bar.find({age:{\$lte:20}})

\$lte#####update()#####service##ID#####

##2#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.ID":{\$lte:15}})

\$ne

##

{<###>#{\$ne:<#>}}

##

\$ne####"###"#####(!=)##"#"####

##

####bar#age#####20####

db.foo.bar.find({age:{\$ne:20}})

\$ne#####update()#####service##type####

##15#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.type":{\$ne:15}})

\$et

##

{<###>#{\$et:<#>}}

##

\$et####"###"#####(=)##"#"####

##

####bar#age#####20####

db.foo.bar.find({age:{\$et:20}}) ### db.foo.bar.find({age:20})

\$et#####update()#####service##type###

##15#####age#####25 #

db.foo.bar.update({\$set:{age:25}}, {"service.type":{\$et:15}})

\$mod

##

##

##

\$in

##

```
{<###>:{$in:[<#1>,<#2>,...<#N>]}}
```

##

```
##### "<###>"#####([<#1>,<#2>,...<#N>])##### "<###>"#####
##### "<###>"#####([<#1>,<#2>,...<#N>])#####
```

##

```
# #####bar#age#####20#25####
```

```
db.foo.bar.find({age:{$in:[20,25]}})
```

```
# $in#####bar#####name####"Tom"#"Mike"#####age####
```

```
db.foo.bar.update({$unset:{age:""},{name:{$in:["Tom","Mike"]}})
```



Note: ##### {<###>:{\$in:[<#>]}}#### {<###>:<#>}

```
db.foo.bar.find(age:{$in:[20]})###db.foo.bar.find({age:20})
```

\$nin

##

```
{<###>:{$nin:[<#1>,<#2>,...<#N>]}}
```

##

```
##### "<###>"#####([<#1>,<#2>,...<#N>])#####
```

```
# "<###>"##### "<###>"#####([<#1>,<#2>,...<#N>])#####
```

##

```
# #####bar#age#####20#25###bar#####age#####
```

```
db.foo.bar.find({age:{$nin:[20,25]}})
```

```
# $nin#####bar#####name#####"Tom"#"Mike"#####bar#####
```

```
###name#####age####
```

```
db.foo.bar.update({$unset:{age:""},{name:{$nin:["Tom","Mike"]}})
```



Note: ##### {<###>:{\$nin:[<#>]}}#### {<###>:{\$ne:<#>}}

```
db.foo.bar.find(age:{$nin:[20]})###db.foo.bar.find({age:{$ne:20}})
```

\$all

##

```
{<###>:{$all:[<#1>,<#2>,...<#N>]}}
```

##

```
$all##### "<###>"#####([<#1>,<#2>,...<#N>])####
```

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##

#####bar#name#####"Tom"#"Mike"####

```
db.foo.bar.find({name:{$all:["Tom","Mike"]}})
```

#####bar##name#####

```
["Tom","Mike",..]
```

```
["Tom","Jhon","Mike",...]
```

#####bar#name#####

```
["Tom","Jhon"]
```



Note:

```
##$all#####db.foo.bar.find(age:{$all:[20]})#####db.foo.bar.find({age:20})
```

\$and

##

```
{$and:[{<###1>}#{<###2>},...,{<###N>}]}
```

##

\$and#####"#"#####(<###1>#<###2>,...,<###N>)####

#####(<###1>)#####false#sequoiadb#####

##

#####bar#age####20#price#####10###

```
db.foo.bar.find({$and:[{age:20},{price:{$lt:10}}]})
```



Note:

Sequoiadb#####and#####(,)#####

```
db.foo.bar.find({age:20,price:{$lt:10}})
```

###and#####{age#{\$lt:20}}and{age:{\$exists:1}}#####\$and#####

#####{age:{\$lt:20,\$exists:1}}#

```
db.foo.bar.update({$inc:{salary:200}},{$and:[{age:{$lt:20}},{age:{$exists:1}}]})
```

```
db.foo.bar.update({$inc:{salary:200}},{age:{$lt:20,$exists:1}})
```

#####bar###age####age####20#####salary

#####200#

\$not

##

```
{$not:[{<###1>},{<###2>},...,{<###N>}]}
```

##

\$not#####"#"#####(<###1><###2>,...,<###N>)####

##

#####bar#age#####20#price#####10###

```
db.foo.bar.find({$not:[{age:20},{price:{$lt:10}}]})
```


\$or

##

```
{ $or: [ {<###1>}, {<###2>}, ..., {<###N>} ] }
```

##

```
$or#####"#"#####(<###1>,<###2>,...,<###N>)#####
```

```
#####true#####
```

##

```
# #####bar#name####"Tom"##age####20#price#####10####
```

```
db.foo.bar.find({name:"Tom",$or:[{age:20},{price:{$lt:10}}]})
```

```
# $or#####age####20#####snapshot##type####"system"#####$inc##
```

```
#####salary####
```

```
db.foo.bar.update({$inc:{salary:200}},{ $or:[{age:{$lt:20}},{ "snapshot.type":"system"}]})
```

\$type

##

```
{<###>:{$type:<BSON type>}}
```

##

```
#####"<###>"#####<BSON type>"###
```

BSON Type

Type	##	#
32-bit integer	#####-2147483648# 2147483647	16
64-bit integer	##### #-9223372036854775808# 9223372036854775807##### #####SequoiaDB### #####	18
double	#####1.7E-308#1.7E+308	1
string	###	2
ObjectID	#####ID	7
boolean	##(true false)	8
date	##(YYYY-MM-DD)	9
timestamp	###(YYYY-MM-DD- HH.mm.ss.ffffff)	17
Binary data	Base64#####	5
Regular expression	#####	11

Type	##	#
Object	##JSON####	3
Array	#####	4
null	#	10

```
##
# #####bar#age#####
db.foo.bar.find({age: {$type:16}})
# #####bar#####content##arr#####
db.foo.bar.find({"content.arr": {$type:4}})

$exists

##
{<###>: {$exists:<0|1>}}

##
##### "<###>"##### "0"##### "<###>"##### "1"##
##### "<###>"####

##
# #####bar#####age###
db.foo.bar.find({age: {$exists:1}})
# #####bar#####content###name#####
db.foo.bar.find({"content.name": {$exists:0}})

$elemMatch

##
{<###>: {$elemMatch:<#>}}

##
##### "<###>"##### "<#>"####

##
# #####
db.foo.bar.find({"add.0": {$elemMatch:"china"}, "add.1": {$elemMatch:"ABC"}})
#####bar###add##### "china"##### "ABC"####
# ##json####
db.foo.bar.find({content: {$elemMatch: {name:"Tom", phone:123}}})
##content###json#####content###name## "Tom" #phone##123####
```

```
$size

##
{"<###>":{"$size:"<#>"}}

##
$size#####"<#>"####

##
# #####bar#####arr####2####
db.foo.bar.find({arr:{$size:2}})

$regex

##
$regex ##### SequoiaDB #### PCRE #####
```

```
$options
$regex#####
# i#####
# m#####pcre#####(^(#####)#####)#####
#####
#####\n#####^$#####
# x#####
#####
# s:#####

##
# #####bar#str#####dh.*fj###
db.foo.bar.find({str:{$regex:'dh.*fj',$options:'i'}})

$options

##

##

##
```

###

###	##	##	##
\$inc	##	#####	db.foo.bar.update({\$inc:{age:25}})

(\$inc:

###	##	##	##
\$set	#####	#####	db.foo.bar.update({\$set:{age:10}})
\$unset	#####	#####	db.foo.bar.update({\$unset:{age:""}})
\$addtoSet	####	##### #####	db.foo.bar.update({\$addtoSet:{array:[3,4,5]}})
\$pop	#####N ###	##### N ##### ##### N ## 0 ##### -N # ### #	db.foo.bar.update({\$pop:{array:2}})
\$pull	####	##### #####	db.foo.bar.update({\$pull:{array:2}})
\$pull_all	####	##### ##### #	db.foo.bar.update({\$pull_all:{array:[2,3,4]}})
\$push	####	##### ### #	db.foo.bar.update({\$push:{array:2}})
\$push_all	####	##### #####	db.foo.bar.update({\$push_all:{array:[2,3,4]}})

```
$inc

##

{$inc:<###1>:<#1>,<###2>:<#2>,...}

##

$inc#####<###>"#####<#>"#####

#####

##

# ####bar#age#####15#####age#####5#ID####1#
db.foo.bar.update({$inc:{age:5,ID:1}}, {age:{$gt:15}})
# ####bar#####arr#####arr#####1#
db.foo.bar.update({$inc:{"arr.1":1}}, {arr:{$exists:1}})

$set

##

{$set:<###1>:<#1>,<###2>:<#2>,...}

##

$set#####<###>"#####<#>"#####

#####
```

##

#####bar####age#####\$set#####

db.foo.bar.update({\$set:{age:5,ID:10}}, {age: {\$exists:0}})

#####bar#####str#####"abc"

db.foo.bar.update({\$set:{str:"abd"}})

##\$set#####arr###bar#####

#####{arr:[1,2,3],name:"Tom"}, {name:"Mike",age:20}#####arr####

db.foo.bar.update({\$set:{"arr.1":4}}, {name: {\$exists:1}})

#####name#####\$set#####arr#####arr###\$set

##arr#####

{arr:[1,4,3],name:"Tom"}, {arr:{"1":4},name:"Mike",age:20}

\$unset

##

{\$unset: {<###1>: "", <###2>: "", ...}}

##

\$unset#####

##

#####bar####name###age#####name#age#####

db.foo.bar.update({\$unset:{name:"",age:""}})

\$unset#####{arr:[1,2,3],name:"Tom"}###\$unset####

#####

db.foo.bar.update({\$unset:{"arr.2":""}})

#####{arr:[1,null,3],name:"Tom"}

\$unset#####{content:{ID:1,type:"system",position:"manager"},name:"Tom"}#

content#####ID#type#position#####\$unset##type#####

db.foo.bar.update({\$unset:{"content.type":""}})

#####{content:{ID:1,position:"manager"},name:"Tom"}

\$rename

##

##

##

\$addtoset

##

{ \$addtoset: {<###1>: [<#1>, <#2>, ..., <#N>] #<###2>: [<#1>, <#2>, ..., <#N>], ... } }

##

\$addtoset#####\$addtoset##

####

#####(<###1>, <###2>, ...)

#####([<#1>, <#2>, ..., <#N>])#####

#####

#####

#####(<###1>, <###2>, ...) #####

#####

##

{arr: [1, 2, 4], age: 10, name: "Tom" }

db.foo.bar.update({ \$addtoset: {arr: [1, 3, 5]} }, {arr: { \$exists: 1 } })

{arr: [1, 2, 4, 3, 5], age: 10, name: "Tom" } #####arr#####3

#5###\$addtoset#####arr####

{name: "Mike", age: 12 }

db.foo.bar.update({ \$addtoset: {arr: [1, 3, 5]} }, {arr: { \$exists: 0 } })

{arr: [1, 3, 5], age: 10, name: "Tom" } #####arr###

\$addtoset###arr#####

\$pop

##

{ \$pop: {<###1>: <N>, <###2>: <N>, ... } }

##

\$pop#####(<###1>, <###2>, ...) ##N#####

#####N#####

##0#####N#<0#### ##### -N ####

##

#####bar#####arr##### {arr: [1, 2, 3, 4], age: 20, name: "Tom" }

db.foo.bar.update({ \$pop: {arr: 2} })

{arr: [1, 2], age: 20, name: "Tom" }

#####bar#####arr###10##### {arr: [1, 2, 3, 4], age: 20, name: "Tom" }

db.foo.bar.update({ \$pop: {arr: 10} })

```

#####{arr:[],age:20,name:"Tom"}
# #####bar#####arr#####N###-2#####{arr:[1,2,3,4],age:20,name:"Tom"}
db.foo.bar.update({$pop:{arr:-2}})

#####{arr:[3,4],age:20,name:"Tom"}

$pull

##

{$pull:{<###1>:<#1>,<###2>:<#2>,...}}

##

$pull#####(<###1>,<###2>,...)# ###(<#1>,<#2>,...) #####
#####
#####

##

# #####bar#####arr##2#####name#####"Tom"#####
{arr:[1,2,4,5],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$pull:{arr:2,name:"Tom"}})

#####{arr:[1,4,5],age:10,name:["Mike"]}
# #####bar#####arr#####2#####name#####"Tom"#####
{arr:[1,3,4,5],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$pull:{arr:2,name:"Tom"}})

#####{arr:[1,3,4,5],age:10,name:["Mike"]}###arr#####2#####
#arr#####

$pull_all

##

{$pull_all:{<###1>:[<#1>,<#2>,...,<#N>],<###2>:[<#1>,<#2>,...,<#N>],...}}

##

$pull_all#####(#<###1>)####([<#1>,<#2>,...,<#N>]) #####
#####

##

# #####bar#####arr###2#3#####name#####"Tom"#####
{arr:[1,2,4,5],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$pull_all:{arr:[2,3],name:["Tom"]}})

#####{arr:[1,4,5],age:10,name:["Mike"]}
# #####bar#####arr#####4#5#####{arr:[1,3,4,5],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$pull_all:{arr:[4,5]}})

```

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```
#####{arr[1,3],age:10,name:["Tom","Mike"]}#

$push

##

{$push:{<###1>:<#1>,<###2>:<#2>,...}}

##

$push #####(<#1>)#####(<###1>)#####
#####
#####

##

# ###bar##arr#####1####arr#####1#####{arr[1,2,4],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$push:{arr:1}})
#####{arr[1,2,4,1],age:10,name:["Tom","Mike"]}#####arr####1###$push#####
#1###arr#####
# ###bar#####name#####{arr[1,2],age:20}
db.foo.bar.update({$push:{name:"Tom"}},{name:{$exits:0}})
#####{arr[1,2],age:20,name:["Tom"]}#####name###$push#####name###
#####

$push_all

##

{$push_all:{<###1>:[<#1>,<#2>,...,<#N>],<###2>:[<#1>,<#2>,...,<#N>],...}}

##

$push_all#####(<###1>)#####([<#1>,<#2>,...,<#N>]) #####
#####([<#1>,<#2>,...,<#N>]) #####
#####

##

# ###bar##arr#####[1,2,8,9]#####{arr[1,2,4,5],age:10,name:["Tom","Mike"]}
db.foo.bar.update({$push_all:{arr:[1,2,8,9]}})
#####{arr[1,2,4,5,1,2,8,9],age:10,name:["Mike"]},#####arr####1#2###$push_all####
##[1,2,8,9]#####arr##
# ###bar#####name#####name#####{arr[1,3,4,5],age:10}
db.foo.bar.update({$push_all:{name:["Tom","Jhon"]}}, {name:{$exits:0}})
#####{arr[1,3,4,5],age:10,name:["Tom","Mike"]}#
```


SQL

Sequoiadb#####SQL#####sequoiaDB#####



Note: sequoiaDB##SQL#####

SQL###

##	##	##
create collectionspace	#####	db.execUpdate("create collectionspace foo")
drop collectionspace	#####	db.execUpdate("drop collectionspace foo")
create collection	####	db.execUpdate("create collection foo.bar")
drop collection	####	db.execUpdate("drop collection foo.bar")
create index	####	db.execUpdate("create index test_index on foo.bar (age)")
drop index	####	db.execUpdate("drop index test_index on foo.bar")
list collectionspaces	#####	db.execUpdate("list collectionspaces")
list collections	####	db.execUpdate("list collections")
insert into	##	db.execUpdate("insert into foo.bar(age,name) values(20,\"Tom\")")
select from	##	db.exec("select * from foo.bar")
update set	##	db.execUpdate("update foo.bar set age=25")
delete from	##	db.execUpdate("delete from foo.bar")
group by	##	db.exec("select dept_no#count(emp_no) as #### from foo.bar group by dept_no ")
group by	##	db.exec("select * from foo.bar order by age desc")
limit	#####	db.exec("select * from foo.bar limit(5)")

##	##	##
offset	#####	db.exec("select * from foo.bar offset(5) ")
as	##	
join on	##	
left outer join on	##	
right outer join on	##	
count	##	
sum	##	
sum	####	
sum	###	
sum	###	

sql create collectionspace

```
create collectionspace ##
#####

##
create collectionspace <cs_name>
<cs_name>#####127B#####

##
# #####foo#####
db.execUpdate("create collectionspace foo") //###db.createCS("foo")
```

sql drop collectionspace

```
drop collectionspace ##
#####

##
drop collectionspace <cs_name>
<cs_name>#####

##
# #####foo#####
db.execUpdate("drop collectionspace foo") //###db.dropCS("foo")
```

sql create collection

```

create collection ##
#####

##
create collection <cs_name>.<cl_name>
<cs_name>#####
<cl_name>#####127B#####

##
# #####foo####bar#
db.execUpdate("create collection foo.bar") //###db.foo.createCL("bar")

```

sql drop collection

```

drop collection ##
#####

##
drop collection <cs_name>.<cl_name>
<cs_name>#####<cl_name>#####

##
# #####foo####bar
db.execUpdate("drop collection foo.bar") //###db.foo.dropCL("bar")

```

sql create index

```

create index ##
#####

##
create [unique] index <index_name> on <cs_name>.<cl_name> (field1_name [asc|desc],...)
[unique]#####
<index_name>####
<cs_name>#####
<cl_name>####
field1_name#####
[asc|desc]####asc#####desc#####

```

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##

#####"test_index"##foo#####bar####age###

```
db.execUpdate("create index test_name on foo.bar (age)")
```

#####desc:

```
db.execUpdate("create index test_name on foo.bar (age desc)")
```

#####

```
db.execUpdate("create index test_name on foo.bar (age desc,name asc)")
```

#

```
db.execUpdate("create unique index test_name on foo.bar (age)")
```

sql drop index

drop index ##

#####

##

```
drop index <index_name> on <cs_name>.<cl_name>
```

<index_name>####

<cs_name>#####

<cl_name>####

##

#####foo###bar###test_index####

```
db.execUpdate("drop index test_index on foo.bar") //###db.foo.bar.dropIndex("test_index")
```

sql list collectionspace

list collectionspace ##

#####

##

```
list collectionspace
```

##

#####

```
db.execUpdate("list collectionspace")
```

###

```
{
  "Name": "testfoo"
  "Name": "big"
  ...
}
```

sql list collections

```
list collections ##
```

```
#####
```

```
##
```

```
list collections
```

```
##
```

```
# #####
```

```
db.execUpdate("list collectionspaces")
```

```
###
```

```
{
  "Name": "testfoo.testbar"
  "Name": "big.small"
  ...
}
```

sql insert into

```
insert into ##
```

```
#####
```

```
##
```

```
insert into <cs_name>.<cl_name>(<field1_name,field2_name,...>) values(<value1,value2,...>)
```

```
##
```

```
insert into <cs_name>.<cl_name> values(<select_set>)
```

```
<cs_name>#####
```

```
<cl_name>####
```

```
<field_name>####
```

```
<value>#####
```

```
<select_set>#####
```

```
##
```

```
# #####bar#####age#name#####(25, "Tom")#
```

```
db.execUpdate("insert into foo.bar(age,name) vaules(25,\"Tom\")")
```

```
# #####bar#####small#####
```

```
db.execUpdate("insert into foo.bar values(select * from big.small)")
```

sql select

```
select ##
```

```
#####
```

```
##
```

```
select * from <cs_name>.<cl_name>
```

```
##
select <field1_name,field2_name,...> from <cs_name>.<cl_name>

<cs_name>#####
<cl_name>####
<field_name>####
```

```
##
```

```
# #####null#
```

```
db.exec("select age,name from foo.bar")
```

```
###
```

```
{
  "age": 10,
  "name": null
}
{
  "age": 10,
  "name": "Tom"
}
}...
```

```
# #####
```

```
db.exec("select * from foo.bar")
```

```
###
```

```
{
  "_id": {
    "$oid": "51c909b0c5b855e029000000"
  },
  "age": 10
}
{
  "_id": {
    "$oid": "51c909b9c5b855e029000001"
  },
  "age": 10,
  "name": "Tom"
}
{
  "_id": {
    "$oid": "51c909c2c5b855e029000002"
  },
  "age": 10,
  "name": "Tom",
  "phone": 123456
}...
```



Note:

1. #####where#group by#order by#limit#offset#####
2. #####(###)


```
###select T.a , T.b from (select * from foo.bar) as T where T.a<10
```
3. #####

sql update

```
update ##
```

```
#####
```

##

```
update <cs_name>.<cl_name> set (<field1_name>=<value1>,...) [where <condition>]
```

<cs_name>#####

<cl_name>####

<condition>#####

##

#####age#####20#####age#####age#20#####

```
db.execUpdate("update foo.bar set age=20")
```

age<10#####

```
db.execUpdate("update foo.bar set age=20 where age<10")
```

sql delete

delete ##

#####

##

```
delete from <cs_name>.<cl_name> [where <condition>]
```

<cs_name>#####

<cl_name>####

<condition>#####

##

#####

```
db.execUpdate("delete from foo.bar")
```

#####age<10####

```
db.execUpdate("delete from foo.bar where age<10")
```

sql group by

group by##

#####

##

```
group by <field1_name> [ASC|DESC ], ...>
```

<field_name>####

[asc|desc]####asc#####desc#####asc

##

#####dept_no###

```
db.exec("select dept_no#count(emp_no) as #### from foo.bar group by dept_no")
```



Note:

#sum#count#min#max#avg#####

sql order by

order by##

#####

##

order by <field1_name [ASC|DESC], ...>

<field_name>####

[asc|desc]####asc####desc####asc

##

#####dept_no#####

db.exec("select dept_no#count(emp_no) as #### from foo.bar group by dept_no order by
dept_no desc")

Note:

#sum#count#min#max#avg#####

sql limit

limit##

#####

##

limit<limit_num>

<limit_num>####

##

#####10####

db.exec("select * from foo.bar limit 10")

sql offset

offset##

#####

##

offset<offset_num>

<offset_num>#####

##

#####5#####5#####

db.exec("select * from foo.bar offset 5")

sql as

as ##

#####(alias)#

##

<cs_name.cl_name | (select_set) | field_name> AS <alias_name>

<cs_name>#####

<cl_name>####

select_set####

field_name####

<alias_name>###

##

####

db.exec("select T1.age,T1.name from foo.bar as T1 where T1.age>10")

####

db.exec("select age as ## from foo.bar where age>10")

#####

db.exec("select T.age,T.name from (select age,name from foo.bar) as T where T.age>10")

sql inner join

inner join ##

#####

##

<collection1_name | (select_set1) as <alias1_name>

inner join

<collection2_name | (select_set2)> as <alias2_name>

[ON condition]

##

#####foo.emp#####foo.dept#####emp_no#####dept_name#

db.exec("select E.emp_no,D.dept_name from foo.emp as E inner join foo.dept as D on
E.dept_no=D.dept_no") Note:

1. #####

select T1.a,T2.b from foo.bar1 as T1 inner join foo.bar2 as T2 on T1.a<10

2. ###join###select * ###

sql left outer join

left outer join ##

left outer join#####(collection1_name)#####(collection2_name)#####

##

```
<collection1_name | (select_set1) as <alias1_name>
left outer join
<collection2_name | (select_set2)> as <alias2_name>
[ON condition]
```

##

#####foo.emp#####foo.dept#####emp_no#####dept_name#

```
db.exec("select E.emp_no,D.dept_name from foo.emp as E left outer join foo.dept as D on
E.dept_no=D.dept_no where D.dept_no<4")
```

sql right outer join

right outer join ##

right outer join#####(collection2_name)#####(collection1_name)#####

##

```
<collection1_name | (select_set1) as <alias1_name>
right outer join
<collection2_name | (select_set2)> as <alias2_name>
[ON condition]
```

##

#####foo.emp#####foo.dept#####emp_no<10#####dept_name#

```
db.exec("select E.emp_no,D.dept_name from foo.emp as E right outer join foo.dept as D on
E.dept_no=D.dept_no where E.emp_no<10")
```

sql sum()

sum() ##

#####

##

```
sum(field_name) as <alisa_name>
```



Note: 1.##sum#####

2.#####

##

###bar#age#####

```
db.exec("select sum(age) as #### from foo.bar")
```

sql count()

```
count() ##
```

```
#####
```

```
##
```

```
count(field_name) as <alisa_name>
```

 Note: 1. ##count#####

```
##
```

```
# ###bar#age#####
```

```
db.exec("select count(age) as ## from foo.bar")
```

sql avg()

```
avg() ##
```

```
#####
```

```
##
```

```
avg(field_name) as <alisa_name>
```

 Note: 1. ##avg#####
2. #####

```
##
```

```
# ###bar#age#####
```

```
db.exec("select avg(age) as #### from foo.bar")
```

sql max()

```
max() ##
```

```
#####
```

```
##
```

```
max(field_name) as <alisa_name>
```

 Note: 1. ##max#####

```
##
```

```
# ###bar#age#####
```

```
db.exec("select max(age) as #### from foo.bar")
```

sql min()

min() ##
#####

min(field_name) as <alisa_name>

👉 Note: 1. ##min#####

###bar#age#####
db.exec("select min(age) as #### from foo.bar")

SQL to sequoiadb ###

#####

SQL	Sequoiadb
database	database
table	collection
row	document / BSON document
column	field
index	index
table joins	embedded documents
primary key #####	primary key #Sequoiadb##primary key#####_id####

Create#Alter

#####SQL#####Sequoiadb#####

SQL ##	Sequoiadb ##	####
create table student (id not null, stu_id varchar(50), age number primary key(id))	#####_id ###_id##### db.collectionspace.student({stu_id:"01",age:20}) ##### db.collectionspace.createCL("student")	insert() , createCL()
alter table student add name varchar(50)	##### #####update()##### \$set#####	update() , \$set

SQL ##	Sequoiadb ##	####
	db.collectionspace.student.update({}, {set:{name:"Tom"}})	
alter table student drop column name	##### #####update()##### \$unset##### db.collectionspace.student.update({}, {unset:{name:"Tom"}})	update() , \$unset
create index index_stu_id on student (stu_id)	db.collectionspace.student.createIndex("index_stu_id", {stu_id:-1})	createIndex("index_stu_id", {stu_id:-1})
drop table student	db.collectionspace.dropCL("student")	dropCL()

Insert

#####SQL#####sequoiadb#####

SQL ##	Sequoiadb ##	####
insert into student(stu_id,age) values("01",20)	db.collectionspace.student.insert({stu_id:"01",age:20})	insert({stu_id:"01",age:20})

Select

#####SQL#####sequoiadb#####

SQL ##	Sequoiadb ##	####
select * from student	db.collectionspace.student.find()	find()
select stu_id,age from student	db.collectionspace.student.find({stu_id:"01",age:20})	find()
select * from student where age>25	db.collectionspace.student.find({age: {\$gt:25}})	find({age: \$gt
select age from student where age=25 and stu_id="01"	db.collectionspace.student.find({age:25,stu_id:"01", {age:25}})	find({age:25,stu_id:"01", {age:25}}
select count(*) from student	db.collectionspace.student.count()	count()
select count(stu_id) from student	db.collectionspace.student.count({stu_id: \$exists {exists:1}})	count({stu_id: \$exists {exists:1}}

Update

#####SQL#####sequoiadb#####

SQL ##	Sequoiadb ##	####
update student set age=25 where stu_id="01"	db.collectionspace.student.update({stu_id:"01", {set:{age:25}})	update({stu_id:"01", {set:{age:25}})

SQL ##	Sequoiadb ##	####
update student set age=age+2 where stu_id="01"	db.collectionspace.student.update({\$inc:{age:2}})	update({data:{id: "01"},

delete

#####SQL#####sequoiadb#####

SQL ##	Sequoiadb ##	####
delete from student where age=20	db.collectionspace.student.remove({age:20})	remove({age:20})
delete from student	db.collectionspace.student.remove()	remove()

##

##

##	##
#####	#####
#####	##BSON###16777168##
###	###\$#####. #

##

##	##
#####	127##
###	###\$###SYS#####. #
#####	#####
#####	4096

####

##	##
#####	127##
#####	###\$###SYS#####. #
#####	4096#8192#16384#32768#65536
#####	##### 64GB#128GB#256GB#512GB#1TB
#####	4096

##

##	##
#####	1024##
#####	##BSON#####-48##
####	#####
#####	1##-1
#####	64

###

##	##
#####	32MB
#####	2GB

##

##	##
#####	7
####	####hostname####IP##
##	#####hostname####
#####	#####

##

##	##
####	#####
###	#####
_id	#####_id#####
####	#####

##

##	##
####	##### #####

Error Code List

Description	Error Code
I0 ##	-1

Description	Error Code
#####	-2
####	-3
#####	-4
#####	-5
#####	-6
####	-7
####	-8
####	-9
####	-10
#####	-11
#####	-12
####	-13
#####	-14
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#####	-28
####	-29
#####	-30
#####	-31
#####	-32
#####	-33
#####	-34
#####	-35

Description	Error Code
#####	-36
#####	-37
#####	-38
#####	-39
#####	-40
#####	-41
#####	-42
#####	-43
#####	-44
#####RID	-45
#####	-46
#####	-47
#####	-48
####	-49
#####	-50
####	-51
#####	-52
#####	-53
#####	-54
#####	-55
\$id#####	-56
#####	-57
#####	-58
#####	-59
#####	-60
####ID	-61
##ID##	-62
#####	-63
#####	-64
#####	-65
####	-66
#####	-67
#####	-68
#####	-69

Description	Error Code
#####	-70
#####	-71
#####	-72
####	-73
#####	-74
####	-75
#####	-76
#####	-77
#####	-78
#####	-79
#####	-80
####	-81
#####	-82
#####	-83
####	-84
#####	-85
#####	-86
#####	-87
#####	-88
#####	-89
#####	-90
#####	-91
#####	-92
#####	-93
#####	-94
#####	-95
####	-96
#####	-97
LSN####	-98
####	-99
#####	-100
####	-101
####	-102
####	-103

Description	Error Code
#####	-104
#####	-105
#####	-106
#####	-107
#####	-108
#####	-109
#####	-110
HTTP#####	-111
####	-112
#####	-113
#####	-114
#####	-115
#####	-116
#####	-117
#####	-118
#####	-119
#####	-120
#####	-121
#####	-122
#####	-123
####	-124
##LSN##	-125
#####	-126
#####	-127
#####	-128
#####	-129
PHP#####	-130
#####	-131
#####	-132
#####	-133
#####	-134
#####	-135
#####	-136
#####	-137

Description	Error Code
DMS#####	-138
#####	-139
#####	-140
sdb#####	-141
sdb#####	-142
sdb#####	-143
sdb#####	-144
sdb#####	-145
####	-146
dms#####	-147
#####	-148
#####	-149
#####	-150
#####	-151
#####	-152
#####	-153
#####	-154
#####	-155
#####	-156
####	-157
#####	-158
#####	-159
#####	-160
#####	-161
#####	-162
\$shard#####	-163
#####	-164
#####	-165
#Group####	-166
Group####	-167
#####	-168
#####	-169
#####	-170
#####	-171

Description	Error Code
#####	-172
#####	-173
#####	-174
#####	-175
#####	-176
#####	-177
#####	-178
#####	-179
####	-180
#####	-181
#####	-182
#####	-183
#####	-184
#####	-185
#####	-186
#####	-187
#####	-188
#####	-189
#####	-190
#####	-191
#####	-192
#####	-193
#####	-194
#####	-195
SQL#####	-196
#####	-197
#####	-198
#####	-199
#####	-200
#####	-201
##json###	-202
#####	-203
#####	-204
#####	-205

Description	Error Code
#####	-206
#####	-207
#####	-208
##Queue###	-209
#####, #####	-210
#####	-211
#####	-212
#####	-213
####	-214
####	-215